



DrillMain Intelligence

Smart drilling insights with next-step guidance and integrated maintenance

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Project Context

An intelligent drilling operations platform that transforms daily drilling report data into actionable insights at segment, well, and fleet levels.

The system integrates report ingestion, KPI dashboards, event severity analysis by depth interval, maintenance risk indicators, fleet-wide reporting and well location mapping.





Problem Statement

Lengthy Reports

Drilling reports are extensive and time-consuming to interpret manually

Delayed Insights

Teams need faster identification of non-productive time and operational issues

Risk Detection

High-risk depth intervals and event severity require clear visualization

Comparative Analysis

Performance comparison across wells and fleet-wide trends are essential

Project Objectives

01

User-Focused Platform

Build intuitive drilling data analysis interface

03

Visual Analytics

Visualize KPIs and depth-segment events clearly

05

Fleet Comparisons

Support fleet-wide comparisons and reporting

02

Data Ingestion

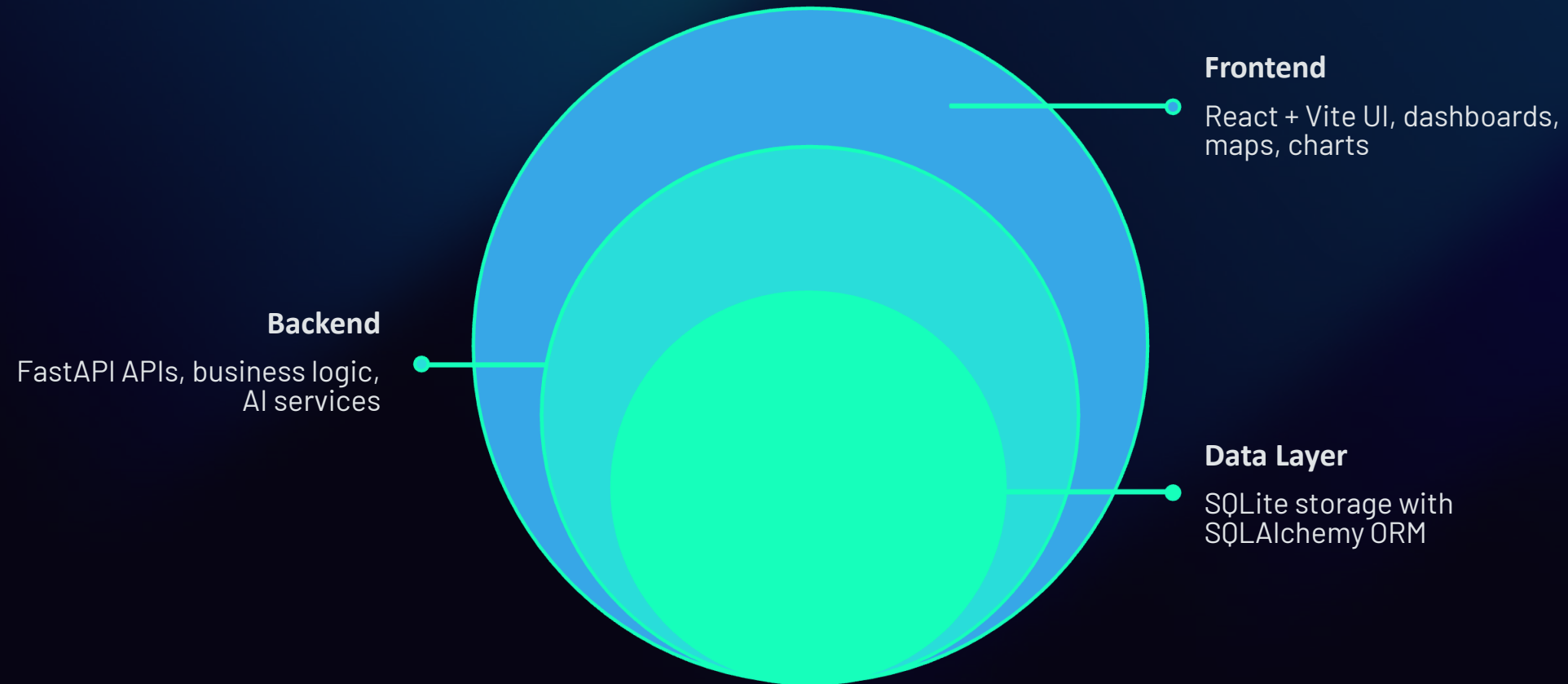
Ingest daily reports and structure operations data

04

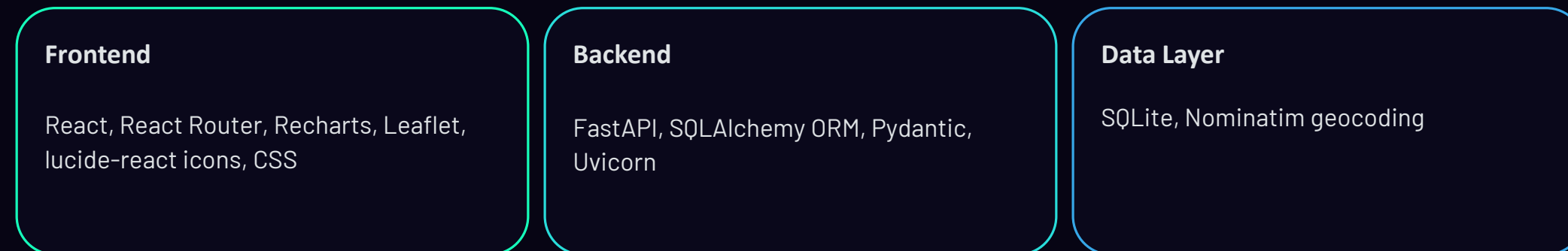
Explainable Analysis

Provide explainable segment-level risk analysis

System Architecture



The architecture is organized in three layers that flow from data storage, through application logic, to the user-facing interface.



Data Processing Flow



Upload

User uploads drilling report file



Ingestion

Backend extracts operations and related fields



Storage

Records stored in database with structured schema



Aggregation

Dashboard endpoints generate KPI metrics



Analysis

Segment analysis produces risk explanations



Fleet Summary

Multi-well aggregation and reporting

Challenges & Solutions



Slow Fleet Loading

Challenge: Heavy per-well calls caused slow fleet/summary loading

Solution: Implemented timeout strategies, non-blocking requests, and lightweight dashboard mode



Generic AI Explanations

Challenge: Repetitive and generic AI-generated explanations

Solution: Developed description-grounded prompts and rule logic tied to actual segment text



Incomplete PDF parsing

Challenge: Missing details from parsed report.

Solution: Integrated editing method to enable users input information to satisfaction



Platform Outcomes



Faster Insights

Rapid interpretation of drilling reports with automated analysis and clear visualizations



Clear Risk Interpretation

Depth-based risk visualization with severity levels and flagged intervals



Cross-Well Visibility

Fleet-wide performance comparison and operational oversight capabilities

Lessons & Future Directions

Key Lessons

- Performance and usability must be designed alongside analytics features
- Explainability is critical for operational trust and adoption
- Small UI decisions significantly impact user adoption rates

Future Enhancements

- Role-based access control and permissions
- Richer report parsing coverage and formats

Project Value: DrillMain Intelligence provides immediate operational value for drilling oversight and risk management, ready for industry application and further research collaboration.

Thank You

We appreciate your time and engagement.

Contact Us

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Next Steps

- Schedule a detailed demo